

Constantinos Menelaou

 Nationality: Greek Cypriot
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 Languages: *English C2, Greek Native*

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SHORT RESUMÉ

2025–Current	Research Assistant UNIVERSITY COLLEGE DUBLIN · Ireland Continuation of the PhD work by studying more cases of boulder transport observed in nature.
2021–2026	PhD Candidate UNIVERSITY COLLEGE DUBLIN · Ireland PhD in Applied Mathematics using Computational Fluid Dynamics to model wave impact on boulders.
2015–2020	Mechanical Engineering with Integrated Masters ARISTOTLE UNIVERSITY OF THESSALONIKI · Greece 5 Year Degree with integrated masters. Specialisation in Fluid Mechanics, Aerodynamics, and Aircraft Design. Thesis on Hypersonic Waverider Aircraft Design.

DEGREES

2025	PhD in Applied Mathematics PHD · UCD
2020	Mechanical Engineer INTEGRATED MASTERS (BENG & MENG) · AUTH

CERTIFICATES & GRANTS

2022	UCD Drone, Data Processing and Visualization Course
2021	MaREI PhD Funding
2021	CFD-FEA Course (A Hands-on Introduction to Engineering Simulations)
2021	English IETLS 2021

TEACHING

UCD Tutor	ICHEC High Performance Computing
UCD Tutor	Numerical Algorithms
UCD Tutor	Statistics with Python
UCD Tutor	Mathematical Fluid Mechanics I
UCD Tutor	Analytical Mechanics
UCD Tutor	PDE for Financial Maths

PROGRAMMING

★★★★★ Python, C/C++, Bash, Linux, Conda, HPC
★★★★★ CUDA, MPI, OpenMP, Rust
☆☆★★★ MATLAB

I have gained experience reading and modifying C++ and CUDA code during my PhD (SPHinXsys and DualSPHysics solvers). DualSPHysics is a C++ SPH solver that utilises the GPU for the computation through CUDA. In DualSPHysics, I extended the runtime gauges by implementing a pressure gauge and moving gauges, which track the motion of a floating body, so that they are fixed in the body's reference frame. Moreover, I implemented two kernel gradient corrections in the particle interaction and non-conservative operator in the momentum equation. SPHinXsys is a C++ SPH library that uses OneTBB for the parallelisation. In SPHinXsys, I couple Project Chrono to take advantage of its collision model. Both forks are on my GitHub. I am also comfortable with OpenFOAM, both the source code (C++) and the solver itself. I am familiar with Windows and Linux (Ubuntu and Arch) for development, with my personal preference being Linux. I have also been using the HPC clusters ICHEC's Key, UCD's Sonic and LuxProvide's Meluxina over the years to run my numerical simulations. I have also tutored for the joint UCD-ICHEC module "ICHEC High Performance Computing" which focused on parallel programming using OpenMP and MPI in C. Furthermore, I am proficient in Python. I have used it for the creation of several tools and packages relevant to my PhD and most of the data analysis. I have also used MATLAB in the past (undergraduate and PhD) whenever Python or other languages were insufficient. Lastly, I am a fan of Rust, which I have used for personal projects such as a gym booking bot with features like a GUI and sound effects.

PUBLICATIONS

- 2026** Displacement and deposition of cliff-bottom rocks using Smoothed Particle Hydrodynamics, *Physics of Fluids* (Accepted, In Press). **Editor's Pick.**
→ Featured in *AIP Scilight*: "Cracking the boulder-tossing mystery" (2026, In Press).
- 2026** Prediction of the displacement by waves of large rocks using smoothed-particle hydrodynamics, *Computers & Fluids*, doi: [10.1016/j.compfluid.2026.107080](https://doi.org/10.1016/j.compfluid.2026.107080).
- 2024** Wave breaking and transport of cliff-top boulders, 18th International SPHERIC Workshop proceedings.
- 2023** Implementation of various-fidelity methods for viscous effects modeling on the design of a waverider, *Aerospace Science and Technology*, doi: [10.1016/j.ast.2023.108141](https://doi.org/10.1016/j.ast.2023.108141).

CONFERENCES

- 2025** 2nd European Fluid Dynamics Conference, 2025, University College Dublin, Dublin, Ireland.
- 2024** Multiphase, Duisburg, Germany.
- 2024** 18th International SPHERIC Workshop, Berlin, Germany.
- 2023** 3rd International Workshop on Waves, Storm Surges, and Coastal Hazards Incorporating the 17th International Waves Workshop, University of Notre Dame, Notre Dame, Indiana, USA.
- 2022** National High-Performance Compute and Data Ecosystem Symposium (HiCoDES), Dublin, Ireland.
- 2022** 6th DualSPHysics Workshop, Universitat Politècnica de Catalunya, Barcelona, Spain.